

The Gap Between Self-Assessed and Actual AI Literacy in Korean University Students.



Course Title	RESEARCH METHOD IN LIBRARY AND INFORMATION (LI3500751-001)		
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Abstract

This study examines differences between self-assessed and actual generative AI literacy among Korean university students and identifies metacognitive errors. The research questions address self-assessment, actual performance, and differences between them. Participants were undergraduate students at Pusan National University, recruited through volunteer and quota sampling. Data were collected using a 5-point Likert-scale self-assessment survey and a 20-item objective test developed by Jin et al. (2025). Data will be analyzed using SPSS with descriptive statistics and paired-samples t-tests will be conducted to compare self-assessed and actual performance. The results identify discrepancies between perceived and actual AI literacy among university students and suggest future research directions for improving AI literacy. The study also provides implications for instructional design integrating AI literacy and metacognitive education.

I. Introduction

1. Problem Statement

Generative AI has become a key learning tool for college students, with about 90% of British students using it for assignment completion and reporting high confidence in their ability to understand and use it (O'Dea et al., 2026). However, a gap may exist between perceived understanding and actual generative AI literacy (Lyu et al., 2024).

2. Significance Statement

Students with limited AI understanding may struggle to use generative AI effectively (Torres, 2025), while those who misjudge their abilities may be less likely to recognize their weaknesses or seek support (Gross & Latham, 2012). Therefore, this study examines metacognitive errors in college students' use of generative AI, providing researchers with directions for future

studies and educators with a basis for designing educational approaches that integrate AI literacy and metacognitive skills.

3. Research Purpose & Questions

This descriptive research examines gaps between Korean university students' self-assessed and tested AI literacy. The research questions are as follows:

RQ1: What are Korean university students' self-assessed AI literacy levels?

RQ2: What scores do Korean university students achieve on a standardized AI literacy test?

RQ3: How do Korean university students' self-assessed AI literacy levels compare with their standardized AI literacy test scores?

II. Literature Review

1. Perception and Understanding of Generative AI

Generative AI is a content-generation technology based on big data and deep learning (Kim, 2025; Park, 2025). AI use among university students ranges from 77% to 90%, primarily for academic tasks like report writing, data analysis, and presentations (Kim, 2025; Lee & You, 2024; Park, 2025). Participants report high satisfaction with its ease and usefulness (Lee & You, 2024).

However, Park (2025) found that students' AI understanding was mostly moderate, with only 5% reporting very high understanding. Despite recognizing source and accuracy issues, students often accepted AI outputs without verification (Kim, 2025; Lee & You, 2024). Lee (2023) found that students needed improvement in digital literacy, particularly problem-solving and IT skills.

2. Impact of AI on Academic Performance

Dunn (2025) found that academic performance after AI adoption varied by task type, with achievement polarization observed. This gap is linked to AI utilization competency (Wang, 2026). Higher levels of critical thinking enhance the positive effects of AI, while lower levels limit learning due to excessive dependency.

Furthermore, Yu (2026) found that while AI increases participation but is less effective than human tutors. Oh et al. (2025) noted that AI use can lead to reliability issues and overdependence. Thus, AI interacts with learner competencies to distort metacognitive judgments.

Conversely, Shin (2025) demonstrated that students actively using generative AI achieved significantly higher learning outcomes than a control group, demonstrating the benefits of structured use.

3. Problem of AI Dependence

AI dependency is habitual reliance that makes independent task completion difficult (Revesai, 2025). Revesai (2025) found that dependent students had GPAs 0.41 points lower, as cognitive tasks were outsourced to AI. Dependency often creates ‘false self-efficacy’, causing anxiety during independent tasks (Jia et al., 2025).

Furthermore, It reduces self-monitoring, triggering ‘metacognitive laziness’ (Fan et al., 2024). This miscalibration aligns with the 'Dunning-Kruger Effect' identified in traditional information literacy studies, where below-proficient users overclaim false knowledge and fail to seek help (Gross & Latham, 2012).

However, psychological factors can reduce this dependence. Lee et al. (2024) found that ‘creative self-belief’ reduces AI dependence through ‘student agency.’

4. Directions for AI Education

AI-era writing education should prioritize thinking processes, strengthen foundational and critical reading skills, emphasize expressive and reflective writing, and evaluate process, verification, and reorganization skills (Lee, 2024; Cho, 2023). Na (2024) argues that evaluation should focus on students' interpretation rather than AI-generated plausibility.

PBL-based AI literacy education has also shown positive effects. Hong & Sa (2025) found that AI-based projects improved technical competence, learning attitudes, and collaboration. Additionally, Kim (2025) developed a rubric for evaluating AI outputs and ethical practice, emphasizing purpose and use over mere tool adoption.

5. Conclusion

Prior research has focused on AI literacy perceptions, outcomes, and educational directions. However, empirical studies on how generative AI affects metacognitive judgment remain scarce. This study addresses this gap by measuring mismatches between self-assessed and actual AI literacy among university students.

III. Methodology

1. Data collection method

To ensure that self-assessed and actual AI literacy are measured under the same conditions, data will be collected through face-to-face surveys and tests. Participants will first complete a self-assessment survey of their AI literacy, followed by a test measuring actual competency. Both instruments are based on the four dimensions of AI literacy proposed by Ng et al. (2021): know & understand, use & apply, evaluate & create, and ethics. The survey uses five-point Likert-scale items adapted from Lee and Park (2024), Lim et al. (2024), Liu and Xiaochong (2025), and Kim (2025), while the test consists of 20 multiple-choice items developed by Jin et al. (2025).

2. The Population of Interest and the Specific Sample to be Researched

The population consists of university students, with Pusan National University students selected as the sample. The study will be conducted in a classroom in the PNU Biology Building. Volunteer and quota sampling based on academic year and major will be used. Participants will be recruited through online and offline channels, and the final sample will be selected according to the predetermined quotas.

3. Operationalized Definition of Key Concept

In this study, AI is limited to generative AI. Generative AI literacy is defined as the ability to critically evaluate, collaborate with, and effectively use AI (Long & Magerko, 2020), and is measured by scores on an AI literacy test. Self-assessment is defined as an individual's subjective evaluation of their own ability and is measured using a five-point Likert-scale survey. Since the maximum scores for the two scales are different, both scores will be converted to percentage scores before being compared.

4. Analytical Approach

This study is a quantitative study that analyzes survey and test results measuring perceived and actual AI literacy. Descriptive statistics, including means and standard deviations, will be calculated using SPSS. A paired-samples t-test will then be conducted to examine differences between the two scores, with statistical significance set at $p < .05$.

5. Rigor and Trustworthiness Strategies

Before data collection, participants will be informed of the study purpose and provide informed consent. Data will be collected anonymously, with only academic year and major recorded. Quota sampling based on academic year and major will be used to enhance sample representativeness and reduce bias toward specific groups.

The self-assessment survey and AI literacy test will be administered and submitted as a single package, ensuring one-to-one matching while maintaining anonymity and supporting data management rigor.

6. Limitations of the Proposed Research Design

This study is limited in its generalizability because the sample consists solely of Pusan National University students. In addition, volunteer sampling may introduce sample bias, as students with greater interest in or experience with generative AI may be more likely to participate.

Consequently, participants' self-assessment and test scores may be higher than those of the broader student population, potentially reducing the observed gap between perceived and actual AI literacy and affecting the validity of the findings. Despite these limitations, the study has academic value in its applicability to comparative research across academic disciplines, particularly for examining differences between perceived and actual AI literacy among students from humanities and social science fields and engineering fields.

7. Ethical Considerations

This study prioritizes participant privacy and voluntary participation as key ethical considerations. Because personal information such as academic year, major, and AI use experience may be collected, participants will be fully informed of the study purpose and data use before providing consent. To protect confidentiality, all data will be anonymized and stored in password-protected files.

In addition, because AI literacy assessments may favor students from certain disciplines or those more familiar with digital technologies, evaluation items will be designed with

participants of diverse backgrounds and skill levels in mind. Through this, bias in the research findings can be reduced and a fairer study can be ensured.

IV. Management Plan

Personnel	Name		Role		Responsibilities			
	Kim Su Bin		Data analyst		• Stats (t-test) & Results			
	Park Ye Won		Project lead		• Sampling & Reporting			
	Seo Yun Ha		Field planner		• Field planning & Promotions			
	Sung Young In		Op manager		• Survey and test execution			
	Cho So Yeon		Lit & PR		• Literature review & Recruiting			
	Ha Heon Eun		Test designer		• Survey and test design			
Schedule (2027~)	Tasks		Jan	Feb	Mar	Apr	May	Jun
	Literature review							
	Test design							
	Sampling & PR							
	Field survey							
	Data analysis							
	Final report							
Dissemination Plans	• Journal of the Korean Society for Library and Information Science • The Journal of Academic Librarianship							
Budget	Item	Cost	Justification					
	Labor	₩120,000	Support for data collection and activity					
	Promotions	₩10,000	Participant Recruiting					
	Printing	₩20,000	Offline Survey Copies					

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Appendices

1. Self-Assessment Survey

GLAT Self-Assessment Survey

Hello,

This survey is being conducted as part of a research project for the Department of Library and Information Science at Pusan National University, titled “*The Gap Between Self-Assessed and Actual AI Literacy in Korean University Students.*” The survey is intended for currently enrolled students at Pusan National University.

The information you provide will be used solely for research purposes and will not be used for any other purpose. All responses will be collected anonymously, and no personal information will be disclosed or shared outside of this research.

There are no right or wrong answers to any of the questions. Please respond honestly based on your own thoughts and experiences. Your sincere participation and complete responses will greatly contribute to the success of this study.

We sincerely appreciate your time and participation in this survey.

1. What is your gender?

① Male ② Female

2. What is your academic year?

① Freshman (1st) ② Sophomore (2nd) ③ Junior (3rd) ④ Senior (4th)

3. What is your major?

4. The following statements concern your ability to use generative AI. Please read each statement and select the response that best reflects your level of agreement.

(1 = Strongly Disagree, 5 = Strongly Agree)

Item		1	2	3	4	5
Know & Understand						
1	I can clearly explain the advantages and disadvantages of generative AI.					
2	I can explain how generative AI systems operate and respond.					
3	I can understand how responses generated by generative AI are produced.					
4	I can understand and interpret concepts and technologies related to natural language processing in generative AI.					
Use & Apply						
5	I can select the generative AI application that best meets my learning needs.					
6	I can use generative AI to generate new ideas or solutions.					
7	I can integrate content provided by generative AI with my own ideas to produce a final outcome.					
8	I can effectively use generative AI for learning, research, presentation preparation, and report writing.					
9	I can design effective prompts for learning with generative AI.					
Evaluate & Create						
10	I can critically evaluate the functions, strengths, and limitations of outputs generated by generative AI applications.					
11	I can determine whether a response generated by generative AI aligns with the intent of the question.					
12	I can identify inconsistencies or contradictions in outputs generated by generative AI.					

13	I verify whether the sources or evidence provided by generative AI are actually reliable.					
14	I can analyze the structure of text or materials generated by generative AI and select only the information I need.					
Ethics						
15	I am familiar with academic ethical principles related to the use of generative AI applications (e.g., plagiarism).					
16	I can analyze the positive and negative impacts of generative AI on individuals and society.					
17	I can identify potential ethical issues associated with the use of generative AI.					
18	I can recognize privacy and information security concerns when using generative AI applications.					

2. Generative AI Literacy Test

Item	Question	Options (Answer Highlighted)
Know & Understand		
1	Which of the following best describes “Generative AI”?	A. AI that creates new content like text, images, or music by learning from existing data. B. An AI system designed to enhance the speed and accuracy of data retrieval in search engines. C. A form of artificial intelligence that focuses on translating languages in real-time. D. AI technology used primarily for managing and organizing large databases.

2	Which of the following statements best describes an LLM (Large Language Model)?	A. It generates text by analyzing and summarizing large volumes of web content. B. It generates text by predicting the next word based on the context of previous words. C. It generates text by translating input text into multiple languages simultaneously. D. It generates text by using pre-defined templates and filling in the blanks.
3	Which of the following tasks can Generative AI perform with a high degree of accuracy?	A. Predicting stock market trends B. Making ethical decisions in complex scenarios C. Diagnosing rare diseases D. Generating human-like text based on prompts
4	In the context of Generative AI, what is "zero-shot learning"?	A. Training a model without any data. B. The ability of a model to perform a task without any task-specific training. C. A method of reducing the model's training time to zero. D. A technique for generating synthetic training data.
5	Which of the following is a potential challenge when using prompt-based development for text generation?	A. The language model can only generate binary outputs. B. The need for extensive labeled data to train the model.

		C. Crafting a prompt that accurately captures the desired context and nuances. D. The requirement for complex feature engineering.
6	What does the term "token" refer to in the context of a large language model (LLM)?	A. A token is a unit of text, such as a word or a subword, that the model processes individually. B. A token is a unique identifier assigned to each user interacting with the language model. C. A token is a security measure used to authenticate API requests to the language model. D. A token is a reward given to users for contributing valuable data to train the language model.
7	Which of the following is NOT a requirement for an AI to be considered artificial general intelligence (AGI)?	A. The ability to learn and adapt to new tasks without human intervention. B. The capability to perform tasks across various domains with human-like proficiency. C. The ability to predict future events with perfect accuracy. D. The capacity to understand and generate natural language.
8	How does RAG (Retrieval-Augmented Generation) enhance the capabilities of an LLM?	A. By improving its grammar and syntax. B. By providing it with real-time and relevant data.

		C. By increasing its computational speed. D. By enabling it to understand multiple languages.
Use & Apply		
9	When using generative AI to create a marketing pitch, which of the following strategies is least likely to be effective?	A. Supplying the AI with information about the target audience B. Asking the AI to include unique selling points and benefits C. Requesting the AI to use persuasive language techniques D. Providing the AI with a list of competitors' products
10	After deploying a customer service chatbot, you notice that it frequently provides outdated information about company policies. What is the best course of action to address this issue?	A. Implement a feedback loop where users can flag outdated information for review. B. Schedule regular updates to the chatbot's training data to include the latest company policies. C. Set up a system where complex or policy-related queries are escalated to human agents for accurate responses. D. Conduct a comprehensive audit of the chatbot's performance metrics to identify areas for improvement.
11	Suppose you have a large dataset of emails and you want to build an application to answer questions based on this dataset. Which of the following scenarios best illustrates the advantage of using RAG over prompting (i.e., without RAG)?	A. You need to generate creative writing pieces based on the email content. B. You want to ensure the model can answer questions even if it has never seen similar questions before.

		C. You need to answer questions that require specific information from different parts of the email dataset. D. You want to reduce the size of the language model to save computational resources.
Evaluate & Create		
12	As a student using a Large Language Model (LLM) to gather information for an assignment, how should you approach the information it provides?	A. The LLM's answers are always more trustworthy than any information you will find on the internet, so you can use them without further verification. B. The LLM's answers are generally more trustworthy than internet sources, but you should still verify the information with other reliable sources. C. The LLM's answers are not necessarily more trustworthy than internet sources, and you should cross-check the information with other credible references. D. The LLM's answers are less trustworthy than internet sources because it relies on outdated information.
13	"It is unlikely for an LLM to provide an accurate summary of the latest financial market trends in real-time." Is this statement true or false?	A. True, because the LLM's data may be outdated due to its knowledge cutoff.

		B. True, because the LLM is not good at handling numbers and structured data. C. False, because the LLM frequently updates its knowledge base. D. False, because the LLM is capable of synthesizing the latest market data automatically.
14	A generative AI tool has provided a summary of a research paper. The summary states, "The study found that increased screen time is directly correlated with decreased attention spans in children aged 8-12." What is your next step?	A. Accept the summary as accurate because AI tools are generally reliable. B. Ask the AI to provide more details about the study's methodology and results. C. Cross-check the summary with the original research paper. D. Use another AI tool to generate a summary for comparison and evaluate the consistency between both summaries
15	While reviewing a video of a well-known public figure making controversial statements, which characteristic confirms the video was not generated by AI?	A. The public figure's voice sounds like themselves. B. The video has a professional and polished appearance. C. The video is high-quality with smooth transitions. D. None of the above.
Ethics		
16	When a generative AI system is used for screening job applications, what issue might	A. The AI system might overlook applicants' unique achievements and extracurricular activities.

	arise concerning the quality and fairness of hiring decisions?	B. The AI system could misinterpret minor formatting differences in resumes. C. The AI system might not effectively handle applications submitted in various languages. D. The AI system could reinforce existing biases found in historical hiring data.
17	In a healthcare startup, an accurate AI model recommends treatments, but doctors don't trust it because they can't understand how the model arrived at its conclusions.	A. The AI model uses obsolete training data. B. The training dataset lacks sufficient diversity. C. The treatment guidelines input are incorrect. D. The AI model behaves as a black box.
18	What are the potential copyright implications for a journalist using an AI-generated image in a commercial article?	A. The journalist needs to check the licensing policy of the AI tool they used. B. The AI-generated image is automatically free to use without any restrictions. C. The journalist must pay a standard licensing fee to use the AI-generated image. D. The image cannot be used in any commercial context because it is AI-generated.
19	Should we impose restrictions on the outputs of generative AI technologies?	A. Yes, to reduce the computational resources required for operating these technologies.

		B. Yes, to prevent the dissemination of harmful or misleading content. C. No, as it would hinder technological innovation and creativity. D. No, because users should have the freedom to access all generated content.
20	Sending personal information to cloud-based generative AI tools has little privacy concerns.	A. True, as this information is encrypted using sophisticated algorithms during the transmission process. B. True, as generative AI tools are black-box systems and cannot output personal information even if it is used for model training. C. False, as generative AI tools train on unencrypted data and can output private information based on their probabilistic nature. D. False, as advancements in quantum computing can easily decipher the encrypted data.

3. Consent Forms

Informed Consent Form

Research Title:

The Gap Between Self-Assessed and Actual Generative AI Literacy In University Students

1. I have read this consent form and fully understand its contents.
2. I have received a full explanation of this study and understand its purpose and procedures.
3. I voluntarily agree to participate in this study.
4. I understand that if I provide Insincere or incomplete responses, I may be excluded from the selection criteria for research participants, and my response data may not be used.
5. I agree that the researchers and supervising institution may review my survey responses and test scores, within the limits of anonymity, for the research.
6. I agree that my responses may be used for statistical analysis to achieve the objectives of this study.
7. I understand that I may refuse to answer or discontinue my participation at any time during the survey, and that I will not face any adverse consequences if I choose to discontinue my participation.
8. I participate in this survey of my own free will and confirm that checking the consent box below has the same effect as signing this consent form.

☐ Yes, I agree to participate in this study.

☐ No, I do not agree to participate in this study.